

Muhammad Abubakar

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PROFESSIONAL SUMMARY

Final-year Cyber Security undergraduate at GIKI with applied research experience in **physical-layer wireless security** (DQN-based artificial noise allocation) and industry exposure across **secure software development** and **QA** at Thingtrax. Project portfolio covers DevSecOps automation, browser-fingerprint privacy, and ML-based intrusion detection — built end-to-end with threat modeling, SAST/SCA gating, and OWASP-grade hardening.

EDUCATION

Ghulam Ishaq Khan Institute (GIKI), Topi, Pakistan — BSc Cyber Security

Sep 2023 – Jun 2027

CGPA: 3.3 / 4.00 | 2x Dean's Honor List (5th and 6th Semester)

Relevant Coursework: Data Structures & Algorithms, Information Security, Information Assurance (NIST / ISO 27001 / GRC), Cybersecurity Principles, Secure Software Design, Mobile & Wireless Security, Secure Design of Algorithms, Computer Networks.

TECHNICAL SKILLS

Security: AppSec, DevSecOps, Network Security, Penetration Testing, Vulnerability Assessment, Incident Response, Threat Modeling, IDS/IPS, Firewalls, OWASP Top 10, SAST/SCA, Zero Trust

Frameworks & Concepts: NIST CSF, ISO 27001, GRC, Secure SDLC, SIEM concepts, SOC workflows, log analysis

Tools: Nmap, Metasploit, Burp Suite, Wireshark, Bandit, pip-audit, GitHub Actions, Cisco Packet Tracer, Prometheus, Grafana

Languages & ML: Python, C++, JavaScript, SQL, Bash, TensorFlow, scikit-learn, CatBoost, Pandas

Systems & Cloud: Linux (Kali, Ubuntu, RHEL), Windows Server

Web & Databases: Flask, React, Next.js, REST APIs, MySQL, PostgreSQL

PROFESSIONAL EXPERIENCE

Thingtrax, Lahore — Software Quality Assurance Intern (Security Testing)

Jun 2025 – Aug 2025

- Designed and executed security-focused test cases across 4 production releases — auth flow validation, input fuzzing, and dependency vulnerability checks alongside functional/regression coverage.
- Triaged defects with developers and helped enforce the team's secure-coding compliance bar; contributed to a digitization workflow that cut document retrieval from ~15 min to under 5 sec.

Thingtrax, Lahore — Secure Software Developer Intern

Jun 2024 – Aug 2024

- Contributed to an industry-sponsored e-commerce platform for Shanghai Traders, implementing core flows (auth, catalog, checkout) with input validation and parameterized queries.
- Coordinated weekly agile syncs for a 10-member team and shipped 50+ high-fidelity UI screens informed by research on 200+ users, improving delivery cadence ~15%.

PROJECTS

DQN-Enhanced Adaptive Artificial Noise Power Allocation — *Wireless Security / Reinforcement Learning* [GitHub](#)

- Modeled an eavesdropper-present MISO wiretap channel (MRT beamforming, artificial noise, Rayleigh fading, imperfect CSI) to study secrecy capacity under realistic conditions.
- Trained a DQN agent to adaptively split power between signal and jamming noise — gained +0.299 bits/s/Hz secrecy rate over fixed allocation at low SNR.

Phantom Identity — *Browser Fingerprint Privacy Extension (JavaScript)*

[GitHub](#)

- Defends against passive browser-fingerprint tracking by spoofing Canvas, WebGL, Navigator, Screen, and timezone attributes per session via a Manifest V3 extension.
- Surfaces fingerprint entropy and spoof effectiveness through an in-extension dashboard; randomization runs locally with zero outbound telemetry.

Secure DevSecOps CI/CD Pipeline — *AppSec Automation*

[GitHub](#)

- Shifted security left by enforcing fail-fast gates in GitHub Actions — pytest, Bandit (SAST), pip-audit (SCA) — blocking any push with secrets, SQLi patterns, or vulnerable deps.
- Packaged the workflow as a reusable template so downstream repos inherit the same security posture without per-project configuration.

Bakri Pay — *Secure Banking Application* — *Flask / AppSec*

[GitHub](#)

- Full-stack Flask (MVC) banking platform with secure session auth, transaction handling, and rules-based fraud detection on suspicious transfers.
- Mapped controls to OWASP Top 10: bcrypt/Argon2 hashing, CSRF tokens, parameterized queries, and authenticated REST endpoints; validated via Burp Suite vulnerability assessment.

AI-Integrated Intrusion Detection System — *Machine Learning / Cybersecurity*

[GitHub](#)

- Hybrid CatBoost + LOF detector combining supervised classification with unsupervised novelty detection to catch unseen attacks; trained on 1.6M samples in ~7.5 min.

CERTIFICATIONS

- Google Cybersecurity Professional Certificate — Coursera (2025) | AI for Cyber Security — Macquarie University (2025)
- Linux Essentials — LPI | Cisco Networking Basics & Introduction to Cybersecurity — Cisco (2025)